

# Sentinel: An Agentic System for Risk Monitoring and Decision Recommendation on Decentralised Financial Networks

Anonymous ACL submission

## Abstract

Decentralised finance (DeFi) exposes supervisors to fast-moving, networked credit risk: shocks propagate across protocols before human operators can inspect raw on-chain data. LLM agents are a natural response, but they are hard to trust with high-stakes interventions—handed weak or stale evidence they can *over-read* it, and prior systemic-risk evaluations give no regulator-aligned way to measure the resulting false alarms. We introduce Sentinel, a forecast-grounded agentic supervision system that routes LLM decisions through structured evidence rather than raw transactions. Forecast-FM, a graph time-series foundation model, forecasts future exposure networks; deterministic monitors and stress scenarios convert the forecasts into typed alerts and scenario losses; data-health and confidence gates decide when intervention-level tickets are admissible; and every released ticket carries its evidence bundle, rationale, and gate states for audit. On Decision-Bench, a six-axis harness with a regulator-aligned absolute-loss ground truth, five years of weekly DeFi exposure graphs, and eight reference implementations, the full stack lifts ticket F1 over a conservative rules baseline from 0.0076 to 0.0233 ( $p < 10^{-4}$ ), at a visible false-intervention cost under an Opus 4.7 decision model (FIR  $\approx 0.44$ ). Swapping in a tier-below Sonnet 4.6 raises F1 to 0.0288, eliminates measured false interventions, and directionally improves explanation quality at  $\sim 5\times$  lower inference cost. These results position Sentinel as an auditable recall–explanation Pareto option for human-in-the-loop DeFi supervision, not a replacement for conservative rules; we release the evaluation harness and ticket-level audit logs from the historical back-

lending protocols, decentralised exchanges, stablecoins, bridges, and yield aggregators. The May 2022 Terra/Luna collapse erased over \$40 billion in 48 hours, and FTX and SVB/USDC followed within months (Vidal-Tomás et al., 2023): shocks traverse the network faster than supervisors can inspect raw on-chain data, motivating decision-support systems that forecast emerging exposure risk and recommend timely supervisory responses.

A straightforward response is an LLM agent over on-chain measurements. Our reference implementations (§6) show the obvious designs fail in two ways. Reading raw on-chain snapshots, the LLM is conservative but weak: poorly targeted tickets (F1 0.0183) and the lowest-scored rationales under every judge we test. Fed richer *forecasted* evidence, it targets better but *hallucinates severity*: every cited number is traceable to the evidence bundle (grounding 1.000), yet 44.8% of its intervention-level tickets point at protocols that are never actually stressed. The second failure is far costlier in a supervisory setting: every intervention is observable and politically expensive.

We introduce Sentinel (§4), a forecast-grounded agentic system for DeFi risk monitoring and decision recommendation that routes LLM decisions through structured evidence and *measures* the over-intervention cost that richer evidence induces. A graph time-series foundation model (Forecast-FM) forecasts future exposure networks; deterministic monitors and stress scenarios convert the forecasts into typed evidence; an LLM emits ranked supervisory tickets with targets, severities, and evidence-linked rationales; and data-health and confidence gates restrict intervention-level recommendations. Every ticket is released with its evidence bundle, rationale, and gate states for auditability.

We evaluate with Decision-Bench (§5), a six-axis harness over eight reference implementations. Its decision axis scores tickets against a regulator-aligned *absolute-loss* ground truth (§3) rather than

## 1 Introduction

Decentralised finance (DeFi) creates fast-moving networks of token-mediated credit exposure among

the fractional-change ranking of prior systemic-risk evaluations (Bertomeu et al., 2024; Gonon et al., 2025; Li et al., 2025), so false interventions are measured directly—a cost that finance-specific LLM-agent work (Yao et al., 2026; Mao et al., 2025; Luo et al., 2025; Jeon and Lee, 2026) has not reported. Across five years of weekly DeFi exposure graphs the full stack lifts ticket F1 from 0.0076 (conservative persistence-rules baseline) to 0.0233 ( $p < 10^{-4}$ ), and three frontier judges across model families (Zheng et al., 2023) all rank it highest on explanation quality—at a visible false-intervention cost under the Opus 4.7 decision LLM (FIR  $\approx 0.44$ ). A tier-below Sonnet 4.6 decision LLM gives the headline operating point: F1 0.0288, measured false interventions zero, explanation quality directionally higher, at  $\sim 5\times$  lower inference cost. Sentinel is thus an auditable recall–explanation Pareto option for human-in-the-loop supervision, not a replacement for conservative rules.

**Honest scope.** All results come from a historical backtest over 283 weekly snapshots of real DefiLlama-sourced exposure data (2020–2025), not a live deployment—a *methods for deployed systems* and *real-world insights* contribution rather than a deployment case study; the released ticket-level audit logs come from this backtest.

## 2 Related Work

**Bench positioning and ground-truth definition.** LLM-agent benchmarks (HELM (Liang et al., 2022), SWE-bench (Jimenez et al., 2024), Agent-Bench (Liu et al., 2024)) score open-ended reasoning, software repair, and generic agent behaviour; temporal-graph benchmarks (TGB (Huang et al., 2023), OGB (Hu et al., 2020)) score structural prediction quality. Neither scores whether an LLM agent’s supervisory decisions match what a regulator would actually prioritise. The closest prior systemic-risk evaluations (Bertomeu et al., 2024; Gonon et al., 2025; Li et al., 2025) rank protocols by *fractional* weight change, which disproportionately surfaces tiny protocols of low systemic relevance. Decision-Bench replaces this with an *absolute-loss* ground truth (§3).

**LLM agents in finance and DeFi.** General agent patterns such as ReAct (Yao et al., 2023) combine reasoning with tool use; FinGPT (Yang et al., 2023) adapts language models to financial data. Within DeFi, agentic systems have so far targeted transac-

tion auditing (Yao et al., 2026), intent mining (Mao et al., 2025), smart-contract verification (Hu et al., 2026; Kong et al., 2026), price manipulation (Liu et al., 2025), and explanation for graph anomalies (Watson et al., 2025); portfolio-oriented agents combine graph encoders with LLMs (Luo et al., 2025; Jeon and Lee, 2026). These systems consume raw transactions or token text, not structured forecasted evidence, and none reports a false-intervention rate against a regulator-aligned ground truth—a gap our characterisation (§6) closes.

**Forecast-grounded LLM agents in other domains.** Pairing a domain forecaster with an LLM decision layer is an emerging deployment pattern. In macroeconomics, ChatGPT-augmented PMI nowcasting (de Bondt and Sun, 2025) and LLM-driven macroeconomic forecasting (Carriero et al., 2025) both feed structured numeric inputs into an LLM that produces interpretable narratives; a BIS primer surveys the wider pattern (Kwon et al., 2024). Time-series foundation models such as Chronos (Ansari et al., 2024), Lag-Llama (Rasul et al., 2024), and TimesFM (Das et al., 2024) make the forecaster reusable across tasks; tabular foundation models extend the same idea to heterogeneous structured data (Hollmann et al., 2025; Ereemeev et al., 2025). None of these forecast→LLM pipelines, to our knowledge, has been scored against a regulator-aligned ground truth for high-stakes financial-network supervision.

## 3 Preliminaries

**Exposure networks.** We model DeFi credit exposure as a temporal weighted directed graph. At week  $t$  the exposure network is  $G_t = (V_t, E_t, W_t)$ , where  $V_t$  is the active protocol set,  $E_t \subseteq V_t \times V_t$  the directed exposure set, and  $W_t[p, q] = w_{pq,t} \geq 0$  the USD exposure of protocol  $p$  to protocol  $q$  (token-mediated construction in §A). The incident exposure mass of protocol  $v$  is  $w_t(v) = \sum_u w_{uv,t} + \sum_u w_{vu,t}$ ; the observed history  $\mathcal{G}_{1:T} = \{G_1, \dots, G_T\}$  carries node covariates  $X_t$ .

**Monitors and stress scenarios.** At decision epoch  $t$ , a probabilistic temporal-graph forecaster outputs  $\mathcal{P}_{t,h} = P(G_{t+h} \mid G_{\leq t}, X_{\leq t})$  for horizons  $h \in \mathcal{H}$ , summarised by a representative predicted graph  $\hat{G}_{t+h}$  and Monte Carlo samples  $\tilde{G}_{t+h}^{(m)} \sim \mathcal{P}_{t,h}$  (§4.1, §A). A monitor is a functional  $\Phi_j : \mathcal{G} \rightarrow \mathbb{R}$  of a graph (e.g. top-1 PageRank, HHI concentration, density); an alert fires when the fore-

casted statistic  $\Phi_j(\hat{G}_{t+h})$  deviates from the mean  $\mu_{j,t}$  of an  $L$ -week rolling baseline by more than  $z$  baseline standard deviations  $\sigma_{j,t}$ . A stress scenario is a graph perturbation  $s : \mathcal{G} \rightarrow \mathcal{G}$  with relative loss  $L(s; G) = 1 - \text{Mass}(G^s) / \text{Mass}(G)$ ,  $\text{Mass}(G) = \sum_{e \in E(G)} w_e$ ; tail risk over the Monte Carlo samples is the worst- $\lambda$ -fraction mean  $\text{CVaR}_\lambda(s, h)$  (§B).

**Supervisory tickets and ground truth.** Each week the system emits ranked tickets  $\tau = (u, Y, r)$ : an action  $u$  from the four-action playbook  $\mathcal{U} = \{\text{MONITOR}, \text{INVESTIGATE}, \text{RECOMMENDREDUCE}, \text{CONTINGENCY}\}$ , a target set  $Y \subseteq V_t$ , and a rationale  $r$ . Ground truth is regulator-aligned *absolute* loss: with  $\Delta_t^h(v) = w_t(v) - w_{t+h}(v)$ , the stressed set  $\mathcal{S}_t^h$  holds the top- $\pi$  fraction of protocols ranked by  $\Delta_t^h(v)$  among  $\{v : w_t(v) > 0, \Delta_t^h(v) > 0\}$ . Ticket precision, recall, and F1 score the union of ticket targets  $\hat{S}_t$  against  $\mathcal{S}_t^h$ . With  $\hat{S}_t^{\text{int}}$  the targets of *intervention-level* actions (RECOMMENDREDUCE, CONTINGENCY), the false-intervention rate  $\text{FIR} = |\hat{S}_t^{\text{int}} \setminus \mathcal{S}_t^h| / |\hat{S}_t^{\text{int}}|$  is the fraction of intervention targets that are never actually stressed.

**Gates.** A data-health score  $DH_t \in [0, 1]$  averages four data-quality sub-scores; each alert carries a confidence  $C_t(a) \in [0, 1]$  that decays with low data health, wide Monte Carlo dispersion, and longer horizons. Intervention-level actions require  $DH_t \geq \tau_{\text{data}}$  and mean active-alert confidence  $\bar{C}_t \geq \tau_{\text{conf}}$  (§B).

## 4 Pipeline

Sentinel is a four-layer pipeline (Figure 1): forecast (Forecast-FM), evidence (monitors, scenarios), LLM decision, and safety gates. It runs at weekly cadence on a single RTX 4090; a decision epoch takes under a minute of GPU time and well under a dollar of LLM API spend (§5).

### 4.1 Forecaster interface

At each decision epoch  $t$  the agent observes the credit-exposure snapshot  $G_t$  and node covariates  $X_t$  (§3). It queries Forecast-FM for a forecast distribution  $\mathcal{P}_{t,h} \leftarrow \text{Forecast-FM.Forecast}(G_t, X_t, h)$  at horizons  $\mathcal{H} = \{1, 4, 8, 12\}$  weeks, constructs an expected-weight predicted graph  $\hat{G}_{t+h}$  by thresholding edge-existence probabilities at  $\pi_{\min} = 0.2$ , and draws  $M = 50$  Monte Carlo samples for uncertainty. Full predicted-graph construction equations are in §A.

### 4.2 Structured evidence

Layer 2 turns  $\hat{G}_{t+h}$  and the MC samples into a typed evidence bundle. Five *monitor functionals* score concentration, fragility, and instability of the predicted graph: top-1 PageRank (Brin and Page, 1998), HHI concentration (Rhoades, 1993), network density, PageRank Gini, and degree Gini. An alert fires when a monitor metric departs from its  $L = 42$  week rolling baseline by more than  $z = 2$  standard deviations; each alert is attached to (i) an attribution top- $K$  ( $K = 5$ ) of contributing edges by marginal contribution, and (ii) an uncertainty-aware confidence score  $C_t(a) \in [0, 1]$  that decays with low data-health, wide MC dispersion, and longer horizons.

In parallel, a *scenario engine* applies five standardised stress shocks—single protocol failure, bridge cluster failure, stablecoin de-peg, sector lending shock, correlated top-10 stress—to both  $\hat{G}_{t+h}$  and each MC sample, and aggregates the worst-half losses as  $\text{CVaR}_{0.5}(s, h)$ . Full monitor and scenario definitions are in §B.

### 4.3 LLM decision layer

Layer 3 receives the evidence bundle as a typed JSON payload—alerts with attribution top- $K$ , scenario CVaR table, MC dispersion, and the scalar data-health score  $DH_t$ —and is asked to emit a ranked list of supervisory tickets. Each ticket carries (a) a severity level drawn from a fixed four-action playbook (Monitor, Investigate, Recommend-Reduce, and Contingency), (b) a target set of protocols, and (c) a free-form rationale that cites specific evidence fields by name. Self-consistency is probed with three repeated calls at temperature 0 (Jaccard overlap on targets and severities).

### 4.4 Safety gates

Two gates determine the feasibility of intervention-level actions (§3). The *data-health gate*  $DH_t < \tau_{\text{data}} = 0.7$  triggers safe mode, in which only Monitor and Investigate tickets are emitted, regardless of LLM recommendation. The *confidence gate*  $\bar{C}_t < \tau_{\text{conf}} = 0.6$  blocks intervention-level tickets even when data is healthy. Every emitted ticket carries the full evidence bundle, rationale, and gate states (§7).

The end-to-end algorithm is in §B (Algorithm 1).

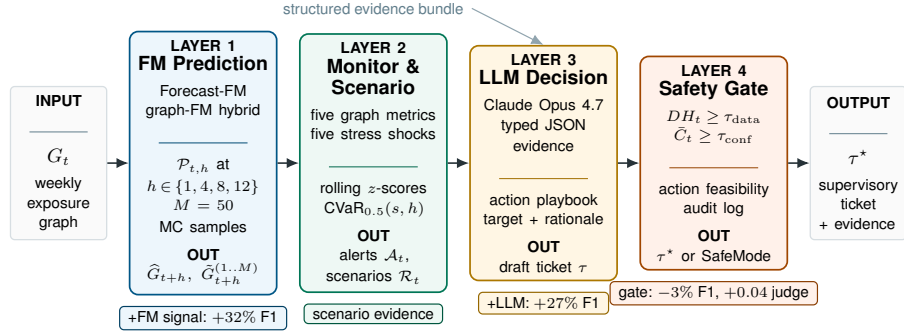


Figure 1: Sentinel: a four-layer pipeline. The agent ingests the weekly exposure graph  $G_t$ , runs Forecast-FM prediction (Layer 1), monitor and scenario engines (Layer 2), an LLM decision layer (Layer 3), and a safety gate (Layer 4) emitting either a supervisory ticket  $\tau^*$  or a safe-mode notice; bottom annotations give per-layer F1 / judge lifts (§6).

## 5 Evaluation Harness

**Why a new harness.** Scoring an LLM-agent supervisory pipeline requires evidence on six independent axes that no existing benchmark covers together: how good is the forecast (b1\_forecast), how early does the monitor turn (b2\_warning), is the predictive uncertainty calibrated (b3\_calibration), do scenario losses match reality (b4\_stress), do the emitted tickets match what a regulator would prioritise (b5\_decision), and what happens under degraded data (b6\_robustness). No existing benchmark covers these axes together (§2); Decision-Bench unifies them on a shared dataset and reference-method set so layer-wise ablations can be reported independently.

**Dataset and split.** We use the Exposure-Corpus dataset of weekly directed weighted exposure graphs (4,300 protocols, 24,300 tokens, 43.7M exposure entries, 283 weekly snapshots from 2020-03 to 2025-08 (Wu et al., 2025)). We adopt a strict no-leakage temporal split: 2020-03–2024-06 train ( $\sim 222$  weeks), 2024-07–2024-12 validation ( $\sim 26$  weeks, used for conformal calibration and threshold tuning), 2025-01–2025-08 test ( $\sim 33$  weeks, frozen leaderboard). All Forecast-FM fine-tuning and model selection likewise use pre-2025 data only (§A). For b2\_warning we additionally run a historical event study around Terra/Luna (2022-05-09), FTX (2022-11-07), and SVB/USDC (2023-03-10); the event study is reported separately from the 2025 predictor-specific leaderboard.

**Regulator-aligned absolute-loss ground truth.** Prior systemic-risk evaluations rank protocols by *fractional* weight change, which silently up-

weights tiny protocols of low systemic relevance. Decision-Bench ranks instead by *absolute* loss, scoring tickets against the stressed set  $\mathcal{S}_t^h$  and FIR of §3. We fix  $\pi = 0.05$  and the ground-truth horizon  $h = 4$  weeks throughout (yielding  $|\mathcal{S}_t^h| \approx 283$  protocols per week); the last four test weeks lack a 4-week-ahead snapshot, so decision axes score  $n = 29$  weeks. The same ground truth anchors b2\_warning, b5\_decision, and the LLM-decision evaluation pipeline.

**Eight reference implementations.** The harness bundles eight reference methods spanning the four ablation cuts of the four-layer framework: heuristic streaming monitor (h1), persistence+rules (m1), snapshot LLM (m2), EvolveGCN (Pareja et al., 2020) (m3), Forecast-FM-only (m4), Forecast-FM+rules (m5), Forecast-FM+LLM (m6), and Forecast-FM+LLM+gate (m7). Each method exposes the same `predict_graph` or `decide_actions` interface so adding a ninth method is a single-file change. The full method specification and per-axis schema are in §C–§D.

**Reproducibility.** A complete leaderboard run finishes in under four GPU-hours on a single RTX 4090 plus approximately \$10 of LLM API spend, making the harness accessible without industrial compute. The harness fixes seeds, snapshots library versions, and hashes checkpoints; every reported number reproduces bit-exactly from the released artefacts.

## 6 Empirical Results

We use Decision-Bench to ask three questions. Does Forecast-FM provide forecast signal that persistence does not (b1, b3, b6)? Does routing the

LLM through structured evidence improve supervisory tickets once precision, recall, explanation quality, and false interventions are scored jointly (b4, b5)? Which components are load-bearing, and which are reserves under degradation (ablations)?

**Forecast-FM contributes trend, calibration, and robustness; persistence still wins on static point error.** On static error metrics, persistence is a deceptively strong baseline: at  $h = 4$  it beats the FM on PageRank MAE ( $3.4 \times 10^{-5}$  vs  $4.5 \times 10^{-5}$ ) and edges it on Spearman rank correlation (0.570 vs 0.558). The FM contributes signal where persistence is structurally pinned: trend consistency 0.525–0.632 across horizons (persistence: 0); b3\_calibration PI coverage 0.913 against a 0.90 target and ECE 0.013; b6\_robustness mean relative degradation 0.113 vs 0.148 for persistence (24% less). The full per-horizon and per-regime audits are in §E.

**Headline decision-quality result.** Table 1 consolidates the central b5\_decision comparison. The persistence+rules baseline (m1) achieves the highest single-axis precision (0.720); like m2/m5 it never escalates beyond Investigate, so its FIR is undefined, not a safety guarantee. Against this baseline, the full Forecast-FM+LLM+gate stack (m7) lifts ticket F1 from 0.0076 to 0.0233 (+207%;  $p < 10^{-4}$ ). Against the comparable pure-LLM baseline that consumes raw snapshots without the forecaster (m2, the only other LLM-bearing row), m7 lifts LLM-as-judge explanation quality from 2.24 to 2.45 on a 1–5 scale (+9.4%) under a tier-above Claude Opus 4.8 judge (Zheng et al., 2023); the lift is significant under GPT-5.5 ( $p < 0.001$ ) but not Opus 4.8 ( $p = 0.23$ ; §F)—directional rather than confirmed. The cost surfaces sharply once m2 is in the table: the raw-snapshot LLM issues *no* interventions, but adding the FM signal (m2→m6) makes it escalate—and 44.8% of those interventions misfire (FIR 0 → 0.448;  $p < 10^{-4}$ ). The safety gate (m6→m7) reduces FIR only marginally (0.448 → 0.437), and its effects on F1 and judge score are within noise (§F). Both LLM variants reach grounding 1.000, so the failure mode is not citation hallucination but *over-reading the forecaster’s evidence as warrant for high-severity action*.<sup>1</sup>

<sup>1</sup>m3\_evolvegc and m4\_fm\_only are predictor-only and do not emit tickets (forecast quality in §E.1); EvolveGCN trails persistence downstream, consistent with its ~220 weekly-snapshot training budget.

**Judge robustness across model families.** We re-judge m2/m6/m7 with a cross-family panel (Table 2). All three frontier judges rank m7 highest (mean score varying by  $\leq 0.45$ ). The one disagreement—Gemini 2.5 Pro scores m2 above m6 (2.69 vs 2.55), still below m7—sits within Gemini’s  $\sim 3\times$  higher weekly noise (std 1.3–1.5 vs 0.3–0.6). Cross-family agreement on the top system closes the judge-family confound.

**A tier-below, cheaper decision LLM dominates the headline operating point.** We re-run the decision layer of m6 and m7 on the *same* FM-derived prompts with two non-Opus 4.7 decision LLMs—Claude Sonnet 4.6 (within-family, tier-below,  $\sim 5\times$  cheaper) and Google Gemini 2.5 Pro (cross-family)—and re-score with the same Opus 4.8 judge (Table 3). On both architectures Sonnet 4.6 is a Pareto improvement over Opus 4.7: F1 rises (+15%/+24%;  $p < 0.001$  on m7), judge quality is directionally higher (+0.11/+0.20; n.s.), and FIR collapses from 0.437–0.448 to 0.000—genuinely: Sonnet 4.6 still issues interventions at Opus-comparable volume ( $\sim 190$  vs 252), all on truly-stressed targets. Gemini 2.5 Pro is more conservative still (lower F1, FIR 0), so over-intervention is a property of the decision model, not of LLM decision layers per se. The operating point a practitioner would now deploy is m7 with Sonnet 4.6 (F1 0.0288, FIR 0.000, judge 2.65).

**Layer-wise contribution.** Holding the decision policy fixed at LLM and adding the Forecast-FM signal lifts ticket F1 by 32% (0.0183 → 0.0241;  $p < 0.001$ ); holding the predictor fixed at Forecast-FM and swapping rules for the LLM lifts F1 by another 27% (0.0190 → 0.0241). The two layers therefore make additive, non-substitutable contributions (Figure 2 in §E). Adding the safety gate trades 3% F1 for judge changes within noise; its value is operational—every intervention clears two gates with auditable logs.

**Matched-budget view.** F1 conflates target quality with volume: m6/m7 emit 6.3–6.5 targets per week vs 4.9 for m2 and 2.1 for m1. Truncating every method to its top- $k$  targets per week ( $k \in \{1, 2, 3\}$ ; §F.3) separates the two effects: the LLM variants still roughly double the rules baseline’s precision@ $k$  ( $p \leq 0.0055$ ), but the FM-fed and raw-snapshot LLM variants become statistically indistinguishable. The FM’s contribution is thus not a sharper top of the ranking but support

| ID | Pred.       | Decision          | Prec.↑       | Recall↑ | F1↑           | Judge↑      | FIR↓  | Ground.↑ | Stabil.↑ |
|----|-------------|-------------------|--------------|---------|---------------|-------------|-------|----------|----------|
| m1 | persist.    | rules             | <b>0.720</b> | 0.004   | 0.0076        | —           | —     | —        | 0.514    |
| m2 | —           | LLM (no FM)       | 0.575        | 0.009   | 0.0183        | 2.24        | —     | 1.000    | 0.532    |
| m5 | Forecast-FM | rules             | 0.600        | 0.010   | 0.0190        | —           | —     | —        | 0.257    |
| m6 | Forecast-FM | LLM               | 0.570        | 0.012   | <b>0.0241</b> | 2.41        | 0.448 | 1.000    | 0.488    |
| m7 | Forecast-FM | LLM + safety gate | 0.580        | 0.012   | 0.0233        | <b>2.45</b> | 0.437 | 1.000    | 0.435    |

Table 1: Headline b5\_decision comparison on the frozen 2025 test split ( $n = 29$ ). Rows are read as a multi-objective trade-off rather than a single leaderboard: m1 is the conservative high-precision baseline; m2 is the pure-LLM baseline (no forecaster); m6/m7 are the FM-grounded LLM variants. Judge scores (1–5 scale) are from a Claude Opus 4.8 judge, strictly tier-above the Claude Opus 4.7 decision LLM; cross-family agreement is in Table 2. — in FIR: undefined— m1/m2/m5 emit no intervention-level ticket that could misfire.

| Method | Opus 4.8    | Gemini 2.5 Pro | GPT-5.5 | Haiku 4.5 <sup>†</sup> |
|--------|-------------|----------------|---------|------------------------|
| m2     | 2.24        | 2.69           | 1.90    | 2.03                   |
| m6     | 2.41        | 2.55           | 2.45    | 2.48                   |
| m7     | <b>2.45</b> | <b>2.90</b>    | 2.45    | 2.69                   |

Table 2: Cross-family judge panel on the same Opus 4.7 decisions ( $n = 29$ ). Gemini and GPT use reasoning = {effort : minimal} so the thinking budget matches the Anthropic baseline. <sup>†</sup> Haiku 4.5 is a tier-below weak-judge reference.

| Method | Decision LLM    | F1↑           | FIR↓         | Judge↑      |
|--------|-----------------|---------------|--------------|-------------|
| m6     | Opus 4.7 (main) | 0.0241        | 0.448        | 2.41        |
| m6     | Sonnet 4.6      | <b>0.0276</b> | <b>0.000</b> | <b>2.52</b> |
| m6     | Gemini 2.5 Pro  | 0.0129        | 0.000        | 2.21        |
| m7     | Opus 4.7 (main) | 0.0233        | 0.437        | 2.45        |
| m7     | Sonnet 4.6      | <b>0.0288</b> | <b>0.000</b> | <b>2.65</b> |
| m7     | Gemini 2.5 Pro  | 0.0139        | 0.000        | 2.38        |

Table 3: Cross-decision-LLM panel on the same FM-derived prompts, judge fixed at Claude Opus 4.8 ( $n = 29$ ). Sonnet 4.6 dominates Opus 4.7 on both architectures. The bold row (m7 × Sonnet 4.6) beats every row in Table 1 on every LLM-bearing axis at  $\sim 5\times$  lower cost.

for a *larger* target set at undiminished precision; under Opus 4.7 the expanded tail is where over-intervention concentrates.

### Ablation: load-bearing vs reserve components.

Under clean 2025 data the confidence gate is load-bearing: removing it raises FIR from 0.000 to 0.429. The scenario engine is load-bearing for target extraction: skipping it collapses ticket precision from 0.600 to 0.000. The data-health gate and multi-horizon diversity are dormant on clean data but activate as reserves under degradation. Under severe feature/edge masking (80–98%) the data-health gate suppresses up to 60% of interventions that would otherwise produce false alarms (FIR range [0.27, 0.60] when disabled vs 0.000

when triggered); multi-horizon monitoring generates  $\sim 4\times$  more alerts in the three crisis windows at unchanged precision. Per-regime ablation audits are in §E.6.

### Early-warning timeline (historical event study).

For the three pre-test crisis windows the shared weighted-degree monitor (h1) achieves median lead times of 4–5 weeks with precision 1.000 at SVB/USDC across all alert budgets  $K \in \{5, 10, 20\}$ ; per-budget details in §E.4. This is historical monitor evidence, not a test-split result.

## 7 Deployment Lessons

Three lessons transfer to other forecast→LLM pipelines.

**L1. Grounding  $\neq$  correctness.** Every LLM variant we test achieves grounding 1.000: every cited number traces to a field of the evidence bundle. The FM-grounded variants still pay FIR = 0.448/0.437 (ungated/gated) under Opus 4.7. The failure is not invented evidence but *over-reading* of weak-but-present evidence as justification for high-severity action. Grounding certifies citations, not safety; only a downstream intervention-realism metric such as FIR does.

**L2. Audit logs are a design artefact, not an output.** Every ticket carries the forecast distribution, monitor scores, scenario CVaRs, rationale, gate states, and seeds that produced it, so a more conservative supervisor can retune  $\tau_{\text{data}}$  or  $\tau_{\text{conf}}$  and rerun the decision policy post-hoc, without re-invoking the FM or the LLM.

**L3. Safety gates are reserves, not always-on filters.** On clean 2025 data the data-health gate is dormant—invisible to a leaderboard without a degradation track—but under masking it suppresses up to 60% of FIR-eligible tickets: evaluate gates on dedicated stress regimes.

## Limitations

**Backtest, not field deployment.** All results come from a historical backtest over 283 weekly snapshots of Exposure-Corpus (2020–2025). We have not run Sentinel as a live weekly cron at a supervisory authority. The ticket-level audit logs we release come from the backtest. Field deployment lessons—operator behaviour, end-to-end latency under real data feeds, regulator-in-the-loop calibration—are out of scope for this paper and remain the most pressing future direction.

**Single domain.** Decision-Bench evaluates a single on-chain risk surface (DeFi inter-protocol credit exposure). Generalisation to NFT lending, perpetual derivatives, or cross-chain bridge networks is untested and will require domain-specific recalibration of the conformal split and the stressed-set percentile  $\pi$ .

**Weekly resolution.** The pipeline operates at weekly granularity, but several DeFi shocks unfold within hours—Terra/Luna erased \$40B in under 48 hours. A sub-hourly production deployment would require denser snapshot ingestion and re-evaluation of conformal calibration on the resulting denser time series. The 4–5 week lead times we report should be read as “how early the weekly monitor turns”, not as end-to-end alert latency.

**Judge thinking-budget is held minimal.** The cross-family judge panel (Claude Opus 4.8, Google Gemini 2.5 Pro, OpenAI GPT-5.5) is run with reasoning effort set to *minimal* so the comparison against the non-reasoning Anthropic baseline is a like-for-like comparison of intrinsic model capability rather than thinking budget. Default reasoning mode on Gemini 2.5 Pro and GPT-5.5 would change the reported scores; we treat the reasoning-on configuration as a distinct evaluation surface and leave it for a follow-up study. Practitioners deploying the judge at default reasoning settings should re-calibrate the score thresholds reported here. Human-judge calibration with practising supervisors remains the most pressing methodological extension.

**Offline decision evaluation.** `b5_decision` scores tickets against a regulator-aligned absolute-loss ground truth, which captures supervisory targeting priority but not the downstream effect of supervisory action—whether the recommended risk reduction would actually have averted conta-

gion. A controlled live evaluation with a regulator in the loop, which Decision-Bench approximates but cannot replace, remains the primary direction we plan to extend the harness in.

## Ethics Statement

**Data provenance.** Exposure-Corpus is built from public on-chain measurements aggregated via DefiLlama (DefiLlama, 2026). No private user data, wallet-level personally-identifying information, or off-chain identity data is used. Protocol-level aggregates are the unit of analysis throughout.

**Decision risk.** Sentinel emits supervisory tickets, not automated trades or automated interventions. The high-severity action types (Recommend-Reduce, Contingency) are gated on a data-health score and a confidence score, and the pipeline emits a safe-mode notice when either gate fails. We characterise the system as decision support for a human supervisor, not as an autonomous risk agent. The measured false-intervention rate ( $\text{FIR} \approx 0.44$  for the Forecast-FM+LLM variant) is reported in the headline results table so that downstream operators can weigh the trade-off against their own risk tolerance before deployment.

**LLM use disclosure.** The decision-layer LLM (`m2`, `m6`, `m7`) is Claude Opus 4.7 in the main configuration; Claude Sonnet 4.6 and Google Gemini 2.5 Pro are reported as decision-layer ablations. The LLM-as-judge for the explanation-quality axis is a three-model cross-family panel: Claude Opus 4.8 (within-family upgrade, strictly tier-above Opus 4.7), Google Gemini 2.5 Pro, and OpenAI GPT-5.5 (cross-family). All judges are run with reasoning effort set to *minimal* so the comparison is controlled for thinking budget. Claude Haiku 4.5 from a prior run is reported in the appendix as a weak-judge baseline. Every model is accessed through the OpenRouter unified API at temperature 0.

**Release.** We will release the harness, the eight reference implementations, and the ticket-level audit logs under a permissive open licence. Model checkpoints will be released subject to dataset-licence compatibility checks.

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## A Forecaster Interface and Notation

### A.1 Credit exposure

Let  $X_G(q)$  be the set of tokens issued by protocol  $q$  and  $X_p(t)$  the set of tokens held by protocol  $p$  at time  $t$ . Protocol  $p$  has a credit exposure to protocol  $q$  at  $t$  if  $X_p(t) \cap X_G(q) \neq \emptyset$  (Wu et al., 2025). This token-mediated dependence generalises counterparty credit risk from traditional finance to DeFi.

### A.2 Network representation

Over an interval  $\tau = t_2 - t_1$ , exposures form a weighted directed graph  $G_\tau = (P_\tau, E_\tau)$ . Node weights are  $w_\tau(p) = \sum_{\sigma \in X_p(t_1) \cap X_p(t_2)} v_{t_2, \sigma}$  where  $v_{t_2, \sigma}$  is the USD value of token  $\sigma$  at  $t_2$ ; edge weights aggregate non-negative token flows from  $p$  to  $q$  mapped to issuer  $q$ , as defined in Wu et al. (2025). We adopt the same construction without modification.

### A.3 Forecast primitive and predicted graph

At each decision epoch  $t$  the agent queries  $\mathcal{P}_{t,h} \leftarrow \text{Forecast-FM.Forecast}(G_t, X_t, h)$ . From  $\mathcal{P}_{t,h}$  we build a representative predicted graph via a thresholded expected-weight operator,

$$\hat{w}_{pq,t+h} := \Pr(e_{pq,t+h} = 1) \cdot \mathbb{E}[w_{pq,t+h} \mid e_{pq,t+h} = 1], \quad (1)$$

$$\hat{E}_{t+h} := \{(p, q) : \Pr(e_{pq,t+h} = 1) > \pi_{\min}\}, \quad (2)$$

with  $\pi_{\min} = 0.2$  (tuned on validation). For uncertainty quantification we draw  $M = 50$  Monte Carlo samples  $\{\tilde{G}_{t+h}^{(m)}\}$  by Bernoulli edge sampling and Gaussian weight perturbation of  $\mathcal{P}_{t,h}$ .

### A.4 Hybrid prediction strategy

In practice Forecast-FM keeps all edges from  $G_t$ , reweights them as  $w'_{pq} = w_{pq} + \Pr(e_{pq,t+h} = 1) \cdot \hat{r}_{pq}$  where  $\hat{r}_{pq}$  is the FM residual, and additionally adds new edges whose existence probability exceeds  $\pi_{\min}$ . This preserves network structure while letting the FM adjust weights and predict new connections.

### A.5 Architecture and training

Forecast-FM is built on the GraphPFN graph-tabular foundation backbone (Hollmann et al., 2025; Ereemeev et al., 2025) with a LiMiX-16M tabular transformer encoder: 12 layers, 4 attention heads, hidden dimension 64. The same backbone serves three heads—edge existence (binary), edge weight (regression), and node-TVL change (regression)—trained jointly with a BCE+MSE composite loss. We fine-tune one checkpoint per horizon  $h \in \{1, 4, 8, 12\}$  for 20 epochs with AdamW (learning rate  $10^{-4}$  for heads,  $10^{-5}$  for the backbone). Fine-tuning draws on Exposure-Corpus snapshots from March 2020 to January 2025: 104 training weeks, 12 validation weeks (used for early stopping), and 8 subsequent held-out weeks for internal evaluation. All data that influences weights or checkpoint selection precedes the frozen 2025 leaderboard split (§C). Full-graph in-context inference fits a single RTX 4090; one weekly forecast across all four horizons completes in well under a minute.

Held-out link-prediction quality is AUPRC = 0.978 / 0.973 / 0.967 / 0.967 and AUROC = 0.996 / 0.995 / 0.993 / 0.993 at  $h = 1 / 4 / 8 / 12$  respectively; a frozen GraphPFN baseline (no fine-tuning) achieves AUPRC 0.936–0.940 at the same horizons, isolating the contribution of domain fine-tuning. These intrinsic forecaster numbers do not appear in the main text because the paper’s contribution is the operating point of the full pipeline, not the FM’s headline accuracy.

## B Pipeline Mathematics

### B.1 Data-health gate

The data-health score combines four deterministic checks of the incoming snapshot:

$$DH_t = \text{mean}(\text{fresh}_t, \text{miss}_t, \text{topo}_t, \text{disc}_t) \in [0, 1]. \quad (3)$$

Safe mode is triggered when  $DH_t < \tau_{\text{data}} = 0.7$ : monitor alerts are still emitted but intervention-type tickets are suppressed.

## B.2 Monitor metrics

| ID | Metric              | Definition                                                |
|----|---------------------|-----------------------------------------------------------|
| N1 | Systemic Importance | $\max_{p \in V} \text{PageRank}(\hat{G}_{t+h})_p$         |
| N2 | HHI Concentration   | $\sum_p (d_p / \sum_q d_q)^2$ , $d_p$ weighted out-degree |
| N3 | Network Density     | $ E_{t+h}  /  V ( V  - 1)$                                |
| N4 | PageRank Gini       | $\text{Gini}(\{\text{PR}_p\}_{p \in V})$                  |
| N5 | Degree Gini         | $\text{Gini}(\{d_p\}_{p \in V})$                          |

PageRank uses power iteration with damping 0.85 and 10 iterations. An alert fires when  $|N_{t,h}^{(j)} - \mu_t^{(j)}| / \max(\sigma_t^{(j)}, \epsilon) > z$  with  $z = 2$ ,  $L = 42$  week rolling baseline,  $\epsilon = 10^{-8}$ . The b2\_warning historical event study uses a shorter 26-week ( $\approx 6$ -month) baseline window.

## B.3 Uncertainty-aware alert confidence

With  $\text{CV}_{t,h}^{(j)}$  the coefficient of variation of metric  $j$  across the  $M$  MC samples (clamped to  $[0, 1]$ ),

$$C_t(a) := DH_t \cdot \frac{1}{1 + \text{CV}_{t,h}^{(j)}} \cdot \frac{1}{1 + \ln(1 + h)} \in [0, 1]. \quad (4)$$

$C_t(a)$  decreases monotonically with lower data health, wider predictive spread, and longer horizons.

## B.4 Scenario engine

| ID | Scenario             | Target               | Shock |
|----|----------------------|----------------------|-------|
| S1 | Single protocol fail | Highest-weight node  | 100%  |
| S2 | Bridge cluster fail  | All bridge nodes     | 100%  |
| S3 | Stablecoin de-peg    | All stablecoin nodes | 50%   |
| S4 | Sector lending shock | All lending nodes    | 30%   |
| S5 | Correlated stress    | Top-10 by weight     | 20%   |

Per-scenario loss  $L_{t,h}(s) = (\sum_e w_e(\hat{G}_{t+h}) - \sum_e w_e(\hat{G}_{t+h}^s)) / \sum_e w_e(\hat{G}_{t+h})$ ; CVaR aggregation averages the worst  $\lceil \lambda M \rceil$  MC-sample losses with  $\lambda = 0.5$ .

## B.5 Decision feasibility and ticket scoring

An intervention-level action  $u$  is feasible iff

$$(\text{SAFE\_MODE}_t = 0) \wedge (\bar{C}_t \geq \tau_{\text{conf}} = 0.6), \quad (5)$$

with  $\bar{C}_t$  the mean alert confidence. Feasible tickets are scored as  $\text{Score}_t(u) = w_u \cdot \bar{C}_t \cdot \text{Imp}_t$ , where  $w_u \in \{0.25, 0.50, 0.75, 1.00\}$  is the playbook severity weight and  $\text{Imp}_t$  is the mean CVaR across scenarios (default 1.0 when no scenarios are available).

## B.6 End-to-end procedure

## C Benchmark Details

### C.1 Six-axis schema

| ID             | Capability tested           | Primary metrics                                                                 |
|----------------|-----------------------------|---------------------------------------------------------------------------------|
| b1_forecast    | Temporal graph prediction   | PageRank/HHI/Gini MAE, trend consistency, Spearman; per $h \in \{1, 4, 8, 12\}$ |
| b2_warning     | Streaming anomaly detection | Precision, recall, F1, lead time, alert stability; per event $\times$ budget    |
| b3_calibration | Predictive uncertainty      | PI coverage at 0.90 target, PI width, ECE, CRPS                                 |
| b4_stress      | What-if scenario fidelity   | Loss MAE, distressed-count MAE, propagation depth MAE, target overlap@ $k$      |
| b5_decision    | Supervisory ticket quality  | Precision, recall@ $k$ , F1, target stability, judge score, FIR                 |
| b6_robustness  | Data quality sensitivity    | Per-benchmark metrics under 5 degradation regimes; relative degradation         |

The six axes are chosen to be (i) *non-overlapping*—a system can excel at b1 while failing b5; (ii) *decomposable along the four-layer framework*, so that ablations on the predictor layer (b1, b3, b6), monitor layer (b2), scenario layer (b4), and decision layer (b5) can be reported independently; and (iii) *individually publishable*, so that downstream work may extend a single axis without re-running the whole suite.

---

**Algorithm 1** Sentinel: one decision epoch.

---

**Require:**  $(G_t, X_t)$ ; horizons  $\mathcal{H}$ ; scenarios  $\mathcal{S}_0$ ; thresholds  $(\tau_{\text{data}}, \tau_{\text{conf}}, z, \pi_{\text{min}})$ ; MC count  $M$   
**Ensure:** Alerts  $\mathcal{A}_t$ ; scenario summary  $\mathcal{R}_t$ ; tickets  $\mathcal{D}_t$   
1:  $DH_t \leftarrow \text{DataHealth}(G_t, X_t)$ ;  $\text{SAFE\_MODE}_t \leftarrow \mathbb{I}\{DH_t < \tau_{\text{data}}\}$   
2: **for all**  $h \in \mathcal{H}$  **do**  
3:    $\mathcal{P}_{t,h} \leftarrow \text{Forecast-FM.Forecast}(G_t, X_t, h)$   
4:    $\hat{G}_{t+h} \leftarrow \text{BuildPredGraph}(\mathcal{P}_{t,h}; \pi_{\text{min}})$   
5:    $\{\tilde{G}_{t+h}^{(m)}\} \leftarrow \text{MCSample}(\mathcal{P}_{t,h})$   
6:   Compute  $\mathbf{N}_{t,h} = \Phi(\hat{G}_{t+h}, X_t)$ , attach rolling stats, derive alerts  $\mathcal{A}_{t,h}$  and confidences  $C_t(\cdot)$   
7:   **for all**  $s \in \mathcal{S}_0$  **do**  
8:     Compute  $L_{t,h}(s), \tilde{L}_{t,h}^{(m)}(s), \text{CVaR}_{0.5}(s, h)$   
9:   **end for**  
10: **end for**  
11:  $\mathcal{D}_t \leftarrow \text{PlaybookDecision}(\mathcal{A}_t, \mathcal{R}_t, DH_t, \text{SAFE\_MODE}_t, \tau_{\text{conf}})$   
12: **return**  $(\mathcal{A}_t, \mathcal{R}_t, \mathcal{D}_t)$ 

---

## C.2 Dataset split

| Split      | Period          | Weeks | Usage                                   |
|------------|-----------------|-------|-----------------------------------------|
| Train      | 2020-03–2024-06 | ~222  | Forecaster training                     |
| Validation | 2024-07–2024-12 | ~26   | Conformal calibration, threshold tuning |
| Test       | 2025-01–2025-08 | ~33   | Frozen leaderboard                      |

## C.3 Ground-truth equation

With  $w_t(v) = \sum_{e \ni v} w_t(e)$  the total weight incident on  $v$ , the regulator-aligned stressed set at horizon  $h$  is

$$\Delta_t^h(v) = w_t(v) - w_{t+h}(v), \quad \mathcal{S}_t^h = \text{top}_\pi \{v : w_t(v) > 0, \Delta_t^h(v) > 0\}, \quad (6)$$

with  $\pi = 0.05$  and ground-truth horizon  $h = 4$  weeks. The final four test weeks lack the 4-week-ahead snapshot  $w_{t+4}$  and are not scored, leaving  $n = 29$  evaluated weeks. This is the single ground-truth definition used by b2, b5, and the LLM-decision evaluation pipeline.

## C.4 Reporting protocol

A submission produces one JSON file per {benchmark, method} pair following the harness schema, plus a single results.json aggregating the suite. The full leaderboard requires b1, b3, b5, b6; b2 and b4 are required only for systems that implement a monitor or scenario layer respectively. The harness fixes random seeds, snapshots library versions, and emits SHA-256 hashes of all checkpoint files.

## D Reference Methods

| ID                   | Predictor       | Policy            | Role                            |
|----------------------|-----------------|-------------------|---------------------------------|
| h1_weighted_degree   | —               | heuristic alert   | Streaming anomaly baseline (b2) |
| m1_persistence_rules | $G_{t+h} = G_t$ | rule engine       | Decision baseline               |
| m2_snapshot_llm      | current snap    | LLM               | LLM-only ablation (no forecast) |
| m3_evolvegcnn        | EvolveGCN       | —                 | Learned-GNN baseline            |
| m4_fm_only           | Forecast-FM     | —                 | FM-only ablation                |
| m5_fm_rules          | Forecast-FM     | rule engine       | FM + rules                      |
| m6_fm_llm            | Forecast-FM     | LLM               | Full FM + LLM stack             |
| m7_fm_llm_gated      | Forecast-FM     | LLM + safety gate | Safety-gated FM + LLM           |

The eight methods expose all four cuts of the four-layer ablation: adding the predictor layer (m1→m5), swapping the decision layer (m5→m6), adding the safety gate (m6→m7), and removing the predictor from the LLM stack (m2→m6). All methods conform to a shared predict\_graph/decide\_actions interface; adding a ninth method is a single-file change.

**LLM model panel and decoding.** All LLM-bearing methods (m2, m6, m7) are evaluated under three decision-layer configurations: Claude Opus 4.7 (main), Claude Sonnet 4.6 (within-family, tier-below ablation), and Google Gemini 2.5 Pro (cross-family ablation). Each runs at temperature 0 and

max\_tokens = 4096; self-consistency is probed with three repeated calls per week and Jaccard overlap on emitted ticket targets and severities. 849 850

The LLM-as-judge for the explanation-quality axis is a three-model panel chosen so the judge tier is at or above the decision tier on the public 2026 Artificial Analysis index: Claude Opus 4.8 (within-family upgrade, strictly tier-above Opus 4.7), Google Gemini 2.5 Pro, and OpenAI GPT-5.5 (cross-family upgrades). The Gemini and GPT judges are reasoning-by-default models; we attach OpenRouter’s reasoning = {effort : minimal} control to both so the judge thinking budget matches the non-reasoning Anthropic baseline. Claude Haiku 4.5 from a pilot run is reported as a weak-judge reference. Every model is served through the OpenRouter unified API; observed cost is on the order of a few US dollars per full sweep of the 29-week test split. 851 852 853 854 855 856 857 858

E Full Per-Slice Results 859

This appendix expands the headline numbers of §6 into the per-slice audits underlying them. 860

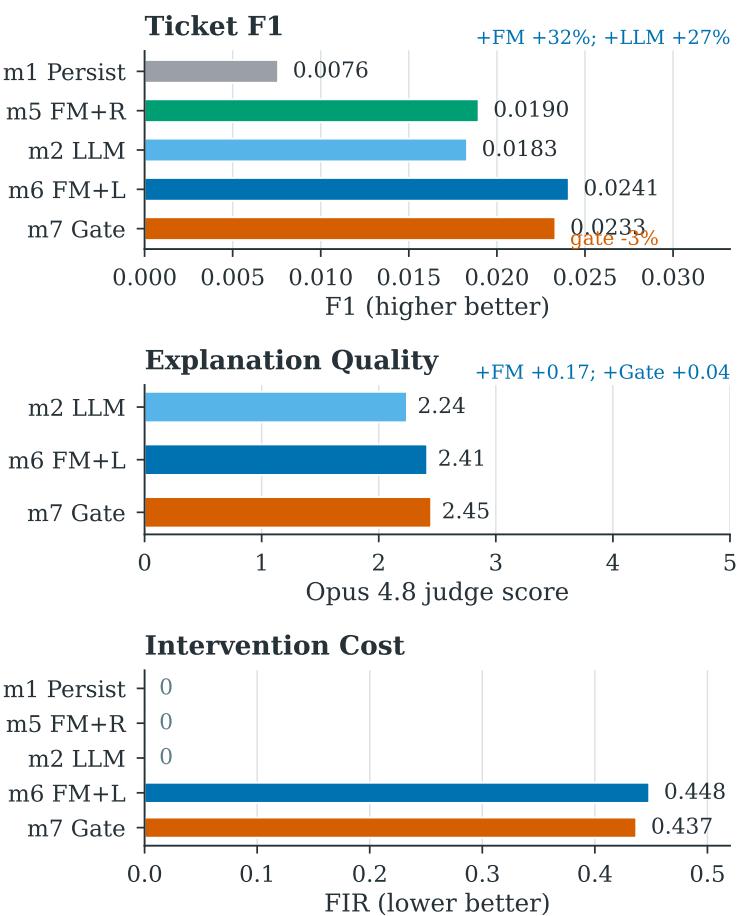


Figure 2: Layer-wise contribution and cost (visualising Tables 1 and 3). Adding the FM signal to a snapshot LLM improves ticket F1 and explanation quality; swapping the FM-fed rule engine for the LLM improves recall-oriented F1 further; the same LLM variants incur a substantial false-intervention rate, partially mitigated by the safety gate. 861 862 863 864 865 866 867 868 869 870

Table 4: b1\_forecast all-horizon forecasting on the 2025 test split. PageRank MAE shown in  $10^{-5}$  units. Best per horizon and metric in **bold**; ties share the bold. Trend consistency for m1\_persistence is structurally 0 because  $\hat{G}_{t+h} = G_t$  produces no trend signal.

| $h$ | Method         | PR MAE ( $\downarrow$ ) | HHI MAE ( $\downarrow$ ) | Gini MAE ( $\downarrow$ ) | Rank Corr ( $\uparrow$ ) | Trend Cons ( $\uparrow$ ) |
|-----|----------------|-------------------------|--------------------------|---------------------------|--------------------------|---------------------------|
| 1   | m1_persistence | <b>3.6</b>              | <b>0.0376</b>            | <b>0.0027</b>             | <b>0.556</b>             | 0.000                     |
|     | m3_evolvegcgn  | 3.7                     | 0.0481                   | 0.1149                    | 0.535                    | 0.444                     |
|     | m4_fm_only     | 4.7                     | <b>0.0376</b>            | 0.0029                    | 0.549                    | <b>0.531</b>              |
|     | m5_fm_rules    | 4.7                     | <b>0.0376</b>            | 0.0029                    | 0.549                    | 0.525                     |
| 4   | m1_persistence | <b>3.4</b>              | 0.0322                   | <b>0.0018</b>             | <b>0.570</b>             | 0.000                     |
|     | m3_evolvegcgn  | 3.5                     | 0.0489                   | 0.1326                    | 0.551                    | 0.324                     |
|     | m4_fm_only     | 4.5                     | <b>0.0321</b>            | 0.0022                    | 0.559                    | 0.621                     |
|     | m5_fm_rules    | 4.5                     | <b>0.0321</b>            | 0.0022                    | 0.558                    | <b>0.628</b>              |
| 8   | m1_persistence | <b>3.9</b>              | 0.0399                   | <b>0.0028</b>             | <b>0.517</b>             | 0.000                     |
|     | m3_evolvegcgn  | 4.0                     | 0.0492                   | 0.1140                    | 0.499                    | 0.296                     |
|     | m4_fm_only     | 4.5                     | <b>0.0398</b>            | 0.0030                    | 0.508                    | 0.624                     |
|     | m5_fm_rules    | 4.5                     | <b>0.0398</b>            | 0.0030                    | 0.508                    | <b>0.632</b>              |
| 12  | m1_persistence | <b>3.7</b>              | 0.0393                   | <b>0.0027</b>             | <b>0.525</b>             | 0.000                     |
|     | m3_evolvegcgn  | 3.8                     | 0.0534                   | 0.1245                    | 0.509                    | 0.314                     |
|     | m4_fm_only     | 4.1                     | <b>0.0392</b>            | 0.0029                    | 0.516                    | <b>0.600</b>              |
|     | m5_fm_rules    | 4.1                     | <b>0.0392</b>            | 0.0029                    | 0.516                    | <b>0.600</b>              |

Table 5: b4\_stress per-scenario detail. Loss MAE in normalised units; Distress count MAE in protocols; target overlap@10 measured against the actual future graph. Best per scenario and metric in **bold**; ties share the bold, and columns where all four methods tie are left unbolded. The main pattern is conservative: FM variants roughly match the strong persistence baseline on loss and target-overlap metrics, while EvolveGCN is generally weaker on downstream what-if fidelity.

| Scenario               | Method         | Loss MAE ( $\downarrow$ ) | Distress MAE ( $\downarrow$ ) | Prop. Depth MAE ( $\downarrow$ ) | Overlap@10 ( $\uparrow$ ) |
|------------------------|----------------|---------------------------|-------------------------------|----------------------------------|---------------------------|
| S1 (single protocol)   | m1_persistence | 0.0668                    | <b>200.3</b>                  | <b>0.897</b>                     | <b>0.314</b>              |
|                        | m3_evolvegcgn  | 0.1072                    | 218.1                         | 0.966                            | 0.034                     |
|                        | m4_fm_only     | <b>0.0667</b>             | 201.4                         | <b>0.897</b>                     | <b>0.314</b>              |
|                        | m5_fm_rules    | <b>0.0667</b>             | 201.6                         | <b>0.897</b>                     | <b>0.314</b>              |
| S2 (bridge cluster)    | m1_persistence | 0.0150                    | <b>27.3</b>                   | 0.000                            | 0.521                     |
|                        | m3_evolvegcgn  | <b>0.0116</b>             | 196.4                         | 0.000                            | 0.189                     |
|                        | m4_fm_only     | 0.0151                    | 50.3                          | 0.000                            | <b>0.525</b>              |
|                        | m5_fm_rules    | 0.0151                    | 52.1                          | 0.000                            | <b>0.525</b>              |
| S3 (stablecoin de-peg) | m1_persistence | <b>0.0079</b>             | 0.0                           | 0.000                            | <b>0.445</b>              |
|                        | m3_evolvegcgn  | 0.0156                    | 0.0                           | 0.000                            | 0.269                     |
|                        | m4_fm_only     | 0.0080                    | 0.0                           | 0.000                            | <b>0.445</b>              |
|                        | m5_fm_rules    | 0.0080                    | 0.0                           | 0.000                            | <b>0.445</b>              |
| S4 (sector lending)    | m1_persistence | 0.0100                    | 0.0                           | 0.034                            | <b>0.536</b>              |
|                        | m3_evolvegcgn  | 0.0200                    | 0.0                           | 0.034                            | 0.293                     |
|                        | m4_fm_only     | <b>0.0099</b>             | 0.0                           | 0.034                            | <b>0.536</b>              |
|                        | m5_fm_rules    | <b>0.0099</b>             | 0.0                           | 0.034                            | <b>0.536</b>              |
| S5 (correlated top-10) | m1_persistence | 0.0100                    | 0.0                           | <b>0.000</b>                     | <b>0.602</b>              |
|                        | m3_evolvegcgn  | 0.0538                    | 0.0                           | 1.000                            | 0.330                     |
|                        | m4_fm_only     | <b>0.0099</b>             | 0.0                           | <b>0.000</b>                     | <b>0.602</b>              |
|                        | m5_fm_rules    | <b>0.0099</b>             | 0.0                           | <b>0.000</b>                     | <b>0.602</b>              |

Table 6: b6\_robustness per-regime detail. All metrics at  $h=4$  on the 2025 test split under five degradation regimes. PageRank, HHI, and Gini MAE in  $10^{-3}$  units. Relative degradation is signed against the clean b1\_forecast baseline; negative values mean the predictor is no worse than on clean data. Best per metric (lowest MAE, highest rank correlation, smallest  $|\Delta_{\text{rel}}|$ ) in **bold**; ties share the bold. Persistence ignores node features, so its noisy\_features\_01 and missing\_features\_20 rows degenerate to the clean numbers (and  $\Delta_{\text{rel}} = 0$ ); EvolveGCN’s predictions also change only negligibly under these two regimes ( $|\Delta_{\text{rel}}| < 10^{-8}$ ).

| Regime              | Method         | PR MAE ( $\downarrow$ ) | HHI MAE ( $\downarrow$ ) | Gini MAE ( $\downarrow$ ) | Rank Corr ( $\uparrow$ ) | $\Delta_{\text{rel}}$ ( $\sim 0$ ) |
|---------------------|----------------|-------------------------|--------------------------|---------------------------|--------------------------|------------------------------------|
| low_data_10pct      | m1_persistence | 15.94                   | 0.0591                   | <b>0.0023</b>             | <b>0.552</b>             | +0.766                             |
|                     | m3_evolvegc    | 15.23                   | 0.0999                   | 0.1301                    | 0.535                    | + <b>0.276</b>                     |
|                     | m4_fm_only     | 12.83                   | <b>0.0586</b>            | 0.0027                    | 0.528                    | +0.612                             |
|                     | m5_fm_rules    | <b>12.71</b>            | 0.0587                   | 0.0027                    | 0.528                    | +0.607                             |
| low_data_25pct      | m1_persistence | <b>9.46</b>             | 0.0282                   | <b>0.0020</b>             | <b>0.577</b>             | −0.093                             |
|                     | m3_evolvegc    | 10.22                   | 0.0486                   | 0.1281                    | 0.558                    | − <b>0.027</b>                     |
|                     | m4_fm_only     | 9.86                    | <b>0.0279</b>            | 0.0024                    | 0.553                    | −0.126                             |
|                     | m5_fm_rules    | 9.95                    | <b>0.0279</b>            | 0.0024                    | 0.553                    | −0.126                             |
| partial_graph_30    | m1_persistence | <b>9.63</b>             | <b>0.0350</b>            | <b>0.0019</b>             | <b>0.554</b>             | +0.068                             |
|                     | m3_evolvegc    | 10.36                   | 0.0488                   | 0.1236                    | 0.537                    | − <b>0.048</b>                     |
|                     | m4_fm_only     | 12.52                   | <b>0.0350</b>            | 0.0022                    | 0.500                    | +0.082                             |
|                     | m5_fm_rules    | 12.45                   | <b>0.0350</b>            | 0.0022                    | 0.499                    | +0.077                             |
| noisy_features_01   | m1_persistence | <b>9.77</b>             | <b>0.0322</b>            | <b>0.0018</b>             | <b>0.584</b>             | <b>0.000</b>                       |
|                     | m3_evolvegc    | 10.68                   | 0.0489                   | 0.1326                    | 0.563                    | <b>0.000</b>                       |
|                     | m4_fm_only     | 11.38                   | 0.0323                   | 0.0021                    | 0.554                    | −0.005                             |
|                     | m5_fm_rules    | 11.35                   | 0.0323                   | 0.0021                    | 0.554                    | −0.007                             |
| missing_features_20 | m1_persistence | <b>9.77</b>             | <b>0.0322</b>            | <b>0.0018</b>             | <b>0.584</b>             | <b>0.000</b>                       |
|                     | m3_evolvegc    | 10.68                   | 0.0489                   | 0.1326                    | 0.563                    | <b>0.000</b>                       |
|                     | m4_fm_only     | 12.11                   | 0.0323                   | 0.0023                    | 0.559                    | +0.017                             |
|                     | m5_fm_rules    | 12.08                   | 0.0323                   | 0.0023                    | 0.559                    | +0.014                             |

Table 7: b2\_warning per-event detail for the shared weighted-degree monitor (h1\_weighted\_degree) across alert budgets  $K \in \{5, 10, 20\}$ . Recall and F1-warning shown in  $10^{-3}$  units (the full DeFi corpus contains  $\sim 4,300$  protocols, so per-week recall denominators are large). Lead time is event-level and not budget-sensitive: 5 weeks for Terra/Luna and FTX, 4 weeks for SVB-USDC. b2\_warning sits upstream of the predictor choice, so a single monitor is shared across all methods (see §6).

| Event      | $K$ | Prec. ( $\uparrow$ ) | Rec. $\times 10^3$ ( $\uparrow$ ) | F1 <sub>w</sub> $\times 10^3$ ( $\uparrow$ ) | Stab. ( $\uparrow$ ) | $ \mathcal{A} $ |
|------------|-----|----------------------|-----------------------------------|----------------------------------------------|----------------------|-----------------|
| Terra/Luna | 5   | 0.60                 | 1.81                              | 3.61                                         | 0.44                 | 11              |
|            | 10  | 0.70                 | 3.26                              | 6.49                                         | 0.62                 | 19              |
|            | 20  | 0.65                 | 7.97                              | 15.74                                        | 0.74                 | 36              |
| FTX        | 5   | 0.60                 | 1.33                              | 2.65                                         | 0.64                 | 10              |
|            | 10  | 0.70                 | 2.39                              | 4.77                                         | 0.72                 | 18              |
|            | 20  | 0.55                 | 4.79                              | 9.49                                         | 0.72                 | 39              |
| SVB-USDC   | 5   | 1.00                 | 1.81                              | 3.62                                         | 0.55                 | 10              |
|            | 10  | 1.00                 | 2.82                              | 5.62                                         | 0.75                 | 15              |
|            | 20  | 1.00                 | 5.64                              | 11.22                                        | 0.81                 | 30              |



Table 8: Per-regime data-health gate ablation isolating A1 from A2. Each row holds  $\tau_{\text{conf}}=0$  and sweeps  $\tau_{\text{data}}$ . “strict” =  $\tau_{\text{data}}=0.85$ , “on” = 0.70, “off” = 0.00. The four regimes apply targeted edge or feature degradation on the clean 2025 test split ( $n_{\text{weeks}}=29$ ): extreme masks both topology and features at 90%, topo removes 90% of edges only, feat corrupts 90% of node features only, severe masks both at 80%. Numbers expand Panel B of Table 9: under degradation the gate-off setting produces FIR up to 0.541, while *on/strict* can suppress intervention tickets via safe-mode and drive FIR to 0.

| Regime       | Gate   | $\tau_{\text{data}}$ | $\overline{d\bar{h}}$ | Safe-mode (%) | Ticket Prec. ( $\uparrow$ ) | FIR ( $\downarrow$ ) | Interventions |
|--------------|--------|----------------------|-----------------------|---------------|-----------------------------|----------------------|---------------|
| extreme      | strict | 0.85                 | 0.587                 | 100           | 0.510                       | <b>0.000</b>         | 0             |
|              | on     | 0.70                 | 0.587                 | 100           | 0.545                       | <b>0.000</b>         | 0             |
|              | off    | 0.00                 | 0.587                 | 0             | 0.490                       | 0.510                | 57            |
| extreme_topo | strict | 0.85                 | 0.637                 | 100           | 0.517                       | <b>0.000</b>         | 0             |
|              | on     | 0.70                 | 0.637                 | 100           | 0.462                       | <b>0.000</b>         | 0             |
|              | off    | 0.00                 | 0.637                 | 0             | 0.559                       | 0.441                | 58            |
| extreme_feat | strict | 0.85                 | 0.755                 | 100           | 0.503                       | <b>0.000</b>         | 0             |
|              | on     | 0.70                 | 0.755                 | 0             | 0.510                       | 0.267                | 5             |
|              | off    | 0.00                 | 0.755                 | 0             | 0.503                       | 0.267                | 5             |
| severe       | strict | 0.85                 | 0.751                 | 100           | 0.490                       | <b>0.000</b>         | 0             |
|              | on     | 0.70                 | 0.751                 | 0             | 0.538                       | 0.400                | 25            |
|              | off    | 0.00                 | 0.751                 | 0             | 0.538                       | 0.541                | 24            |

## E.1 Full all-horizon b1\_forecast results

## E.2 Per-scenario b4\_stress results

## E.3 Per-regime b6\_robustness results

## E.4 Per-budget b2\_warning results

## E.5 Per-regime A1 (data-health gate) isolation

## E.6 Stress-condition ablation Panels B and C

Panel B: under severe feature/edge masking, FIR ranges over  $[0.27, 0.60]$  when the data-health gate is disabled ( $\tau_{\text{data}} = 0$ ; the confidence gate is held off at  $\tau_{\text{conf}} = 0$  throughout Panel B to isolate A1), and collapses to 0.000 when the gate is triggered, by suppressing intervention tickets. Panel C: multi-horizon monitoring ( $h \in \{1, 4, 8, 12\}$ ) generates 73–98 alerts in the three crisis windows vs 19–23 for single-horizon ( $h = 4$ ), at unchanged precision range 0.533–0.624 and zero false interventions in both cases.

## F Statistical Significance and Matched-Budget Analysis

All headline metrics are means over the  $n = 29$  scored test weeks. This appendix reports week-level uncertainty for the central comparisons: 95% percentile bootstrap confidence intervals (10,000 resamples of weeks with replacement; F1 is recomputed from the resampled mean precision and recall) and two-sided paired sign-flip permutation tests (20,000 permutations, methods aligned by week). Analysis code is released with the harness (scripts/bootstrap\_stats.py, scripts/matched\_budget.py).

### F.1 Confidence intervals

### F.2 Paired permutation tests

Three readings follow. (i) The load-bearing quantitative claims—the FM signal’s F1 lift, the full stack’s F1 lift over rules, the Sonnet 4.6 F1 improvement, and the over-intervention cost—are significant at  $p < 0.001$ . (ii) The explanation-quality lift of the FM-grounded variants is judge-dependent: significant under GPT-5.5, positive but not significant under Opus 4.8 ( $p = 0.23$ ), and absent under the (noisiest) Gemini judge. We therefore read the judge axis as directional evidence rather than a confirmed effect. (iii) The safety gate’s effects on F1 and judge score are indistinguishable from noise on clean data, consistent with its characterisation as a reserve mechanism (§7).

Table 9: Component-level ablation of Sentinel on b5\_decision. Panel A shows clean 2025 test-split ablations against the m5\_fm\_rules baseline; Panels B–C report targeted stress runs for mechanisms that are dormant on clean data. A2 (confidence gating) and A3 (scenario engine) are load-bearing in the clean setting, while A1 (data-health gating) and A6 (multi-horizon monitoring) are safety-reserve mechanisms whose effects appear under data degradation and crisis-window dynamics, respectively.

| Panel            | Ablation   | Configuration                                          | Ticket Prec. | FIR       | Notes                               |
|------------------|------------|--------------------------------------------------------|--------------|-----------|-------------------------------------|
| A: clean test    | —          | full m5_fm_rules                                       | 0.600        | 0.000     | baseline                            |
|                  | A1         | disable data-health gate<br>( $\tau_{\text{data}}=0$ ) | 0.600        | 0.000     | no effect (clean data)              |
|                  | A2         | disable confidence gate<br>( $\tau_{\text{conf}}=0$ )  | 0.600        | 0.429     | FIR shoots up                       |
|                  | A3         | skip scenario engine                                   | 0.000        | 0.000     | target extraction collapses         |
|                  | A6         | single-horizon forecasting ( $h=4$ )                   | 0.600        | 0.000     | no effect (clean data)              |
| B: degraded data | A1 off     | 80–98% feature/edge mask,<br>$\tau_{\text{conf}}=0$    | —            | 0.27–0.60 | false interventions emerge          |
|                  | A1 on      | same degradation, safe-mode<br>triggered               | —            | 0.000     | intervention suppression            |
| C: crisis period | A6: single | horizons = {4}, crisis windows                         | 0.533–0.624  | 0.000     | alerts: 19–23                       |
|                  | A6: multi  | horizons = {1, 4, 8, 12}, same<br>windows              | 0.533–0.624  | 0.000     | alerts: 73–98 ( $\approx 4\times$ ) |

Table 10: 95% bootstrap CIs for the decision-quality headline numbers (decision LLM in parentheses; judge is Claude Opus 4.8 throughout).  $^{\ddagger}$ m1’s per-week distribution comes from a local re-run of the released harness: its aggregate F1 (0.0081) brackets the published GPU-run value (0.0076), and the recall@ $k$  and stressed-pool size reproduce exactly, but with  $<1$  ticket per week a few borderline tickets flip across environments (re-run precision 0.633 vs published 0.720), so the m1 interval should be read as order-of-magnitude.

| Configuration          | F1 [95% CI]             | FIR [95% CI]         | Judge [95% CI]    |
|------------------------|-------------------------|----------------------|-------------------|
| m1 rules $^{\ddagger}$ | 0.0081 [0.0041, 0.0123] | 0.000                | —                 |
| m2 (Opus 4.7)          | 0.0184 [0.0142, 0.0223] | 0.000                | 2.24 [2.10, 2.41] |
| m6 (Opus 4.7)          | 0.0241 [0.0201, 0.0281] | 0.448 [0.293, 0.601] | 2.41 [2.21, 2.62] |
| m7 (Opus 4.7)          | 0.0234 [0.0194, 0.0271] | 0.437 [0.282, 0.592] | 2.45 [2.24, 2.66] |
| m6 (Sonnet 4.6)        | 0.0276 [0.0225, 0.0324] | 0.000                | 2.52 [2.28, 2.79] |
| m7 (Sonnet 4.6)        | 0.0288 [0.0236, 0.0335] | 0.000                | 2.66 [2.41, 2.90] |
| m6 (Gemini 2.5 Pro)    | 0.0129 [0.0103, 0.0155] | 0.000                | 2.21              |
| m7 (Gemini 2.5 Pro)    | 0.0139 [0.0111, 0.0167] | 0.000                | 2.38              |

Table 11: Two-sided paired permutation  $p$ -values for the key comparisons. Significant effects ( $p < 0.05$ ) in **bold**.

| Comparison                             | Metric | $p$               |
|----------------------------------------|--------|-------------------|
| m6 vs m2 (FM signal)                   | F1     | <b>0.0002</b>     |
| m7 vs m1 (full stack)                  | F1     | <b>&lt;0.0001</b> |
| m7 vs m6 (gate)                        | F1     | 0.41              |
| m7 $\times$ Sonnet vs m7 $\times$ Opus | F1     | <b>0.0004</b>     |
| m6 vs m2                               | FIR    | <b>&lt;0.0001</b> |
| m6 vs m2 (Opus 4.8 judge)              | Judge  | 0.23              |
| m6 vs m2 (GPT-5.5 judge)               | Judge  | <b>0.0002</b>     |
| m6 vs m2 (Gemini judge)                | Judge  | 0.77              |
| m7 vs m6 (Opus 4.8 judge)              | Judge  | 1.00              |
| m7 $\times$ Sonnet vs m7 $\times$ Opus | Judge  | 0.21              |

### F.3 Matched-budget recall@ $k$

Methods emit different numbers of ticket targets per week (m1 2.1, m2 4.9, m7 6.3, m6 6.5, m7  $\times$  Sonnet 7.6), and unbudgeted recall mechanically favours methods that flag more protocols. To separate target *quality* from target *volume*, we truncate every method to its top- $k$  targets per week (LLM variants ranked by their own emitted risk score; m1 by ticket confidence) and score against the same stressed pool.

The matched-budget view decomposes the headline F1 result into two distinct effects. Against the *rules baseline*, LLM targeting is genuinely better per target: at every  $k$ , precision@ $k$  roughly doubles

Table 12: Matched-budget recall@ $k$  ( $\times 10^3$ ) and precision@ $k$  on the 2025 test split. At matched budgets every LLM variant beats the rules baseline ( $p \leq 0.0055$  vs m1 at all  $k$ ), while the FM-grounded and raw-snapshot LLM variants are statistically indistinguishable from each other (all pairwise  $p \geq 0.10$ ) and the top- $k$  hit patterns of the three FM-fed decision LLMs nearly coincide. m1 row from the local re-run (Table 10 note).

| Method          | $k=1$ |      | $k=2$ |      | $k=3$ |      |
|-----------------|-------|------|-------|------|-------|------|
|                 | R@1   | P@1  | R@2   | P@2  | R@3   | P@3  |
| m1 rules        | 0.71  | 0.24 | 1.71  | 0.28 | 2.45  | 0.26 |
| m2 (Opus 4.7)   | 1.53  | 0.52 | 3.47  | 0.55 | 5.70  | 0.59 |
| m6 (Opus 4.7)   | 1.64  | 0.55 | 3.29  | 0.53 | 5.30  | 0.56 |
| m7 (Opus 4.7)   | 1.64  | 0.55 | 3.47  | 0.55 | 5.30  | 0.56 |
| m7 (Sonnet 4.6) | 1.64  | 0.55 | 3.47  | 0.55 | 5.30  | 0.56 |

(0.52–0.59 vs 0.24–0.28) and recall@ $k$  is significantly higher ( $p \leq 0.0055$ ). Against the *raw-snapshot LLM*, however, the FM’s contribution is not a higher per-target hit rate at the top of the ranking—at matched  $k$  the FM-fed and raw-snapshot variants are statistically indistinguishable, and the FM-fed variants’ top- $k$  targets are nearly identical across decision LLMs because the evidence bundle pins the highest-exposure protocols. The FM’s F1 gain comes from supporting a *larger* target set at undiminished precision (Table 1), which is also where the over-intervention risk concentrates: under Opus 4.7 the expanded tail is escalated to intervention-level severity (FIR = 0.448), while Sonnet 4.6 expands the same set without escalating it (FIR = 0).